

INTERNAL

# Deployment Guide

Quenyx vOPS HUB — v1.0.0 RC1 · Document v2.0

|                  |                            |
|------------------|----------------------------|
| Document Version | 2.0                        |
| Software Version | v1.0.0 RC1                 |
| Applies To       | Quenyx vOPS HUB v1.0.0 RC1 |
| Classification   | Internal                   |
| Owner            | DevOps / SRE               |
| Status           | Released                   |
| Last Updated     | 2026-06-29                 |
| Document Type    | Deployment guide           |

## REVISION HISTORY

| Version | Date       | Notes                                                                                                  |
|---------|------------|--------------------------------------------------------------------------------------------------------|
| 1.0     | 2026       | Initial v1 pack (through Sprint 19).                                                                   |
| 2.0     | 2026-06-29 | Aligned to v1.0.0 RC1; native monitoring scheduler (`observe:run-checks` / `observe:evaluate-alerts`). |

# Table of Contents

- 1. Prerequisites
- 2. Ubuntu deployment (single node)
- 3. Laravel backend
- 4. React frontend
- 5. Node gateway
- 6. Nginx + SSL
- 7. Environment variables (key ones)
- 8. Queues
- 9. Scheduler
- 10. Cache
- 11. AI / RAG / Copilot environment (opt-in)
- 12. Storage
- 13. Database & migrations
- 14. Seeders & corpus
- 15. Backup & restore
- 16. Rollback
- 17. Health checks
- 18. Production checklist

# 10 — Deployment Guide

**Audience:** DevOps, implementation partners. **Canonical source:** This consolidates the repo-root `DEPLOYMENT.md` and adds QCIF/AI specifics. Where the two differ, the root `DEPLOYMENT.md` and the actual `.env.example` prevail. **Host-specific values below are examples — substitute your own.**

## 1. Prerequisites

| Component     | Version                                                                                                                                                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PHP           | <b>8.3</b> recommended (8.1+ supported); extensions: <code>mbstring</code> , <code>pdo_mysql</code> , <code>openssl</code> , <code>curl</code> , <code>fileinfo</code> , <code>bcmath</code> |
| Composer      | 2.x                                                                                                                                                                                          |
| Node.js / npm | 18+ LTS or 20+ / npm 9+ ( <code>npm ci</code> )                                                                                                                                              |
| MySQL         | 8.0+                                                                                                                                                                                         |
| Nginx         | reverse proxy + static frontend                                                                                                                                                              |

## 2. Ubuntu deployment (single node)

Single host runs: Laravel backend (PHP-FPM or `artisan serve`), Node gateway, Nginx (serves `frontend/dist`, proxies `/api/`), MySQL, queue worker, and the Laravel scheduler.

```
git clone <repo-url> && cd quenyx-saas
mysql -u root -p < scripts/mysql-quenyx-setup.sql # first-time DB setup
```

## 3. Laravel backend

```
cd backend
composer install --no-dev --optimize-autoloader
cp .env.example .env
# Edit DB_*, APP_URL, APP_KEY, SEED_ADMIN_PASSWORD, GATEWAY_INTERNAL_SECRET, AI/RAG vars (see §1)
php artisan key:generate
php artisan migrate --force
php artisan db:seed --force
php artisan config:cache && php artisan route:cache && php artisan view:cache
```

## 4. React frontend

```
cd ../frontend
npm ci
# set VITE_API_BASE_URL to your API base (same-origin /api recommended)
npm run build      # output: frontend/dist
```

## 5. Node gateway

```
cd ../gateway
npm ci
npm run build      # rebuild + restart after every change
# env: GATEWAY_PORT=4000, BACKEND_BASE_URL=http://127.0.0.1:8000, ENTITLEMENTS_CACHE_TTL_MS=3000
#      GATEWAY_INTERNAL_SECRET=<strong shared secret, also set in backend>
```

## 6. Nginx + SSL

- Document root → `frontend/dist` ; SPA fallback `try_files $uri /index.html` .
- `location /api/` → `proxy_pass http://127.0.0.1:4000` (gateway), forwarding `Authorization` .
- **SSL**: terminate TLS at Nginx (or LB); redirect HTTP→HTTPS. (See root `DEPLOYMENT.md` for full server blocks, single-node and multi-node.)

## 7. Environment variables (key ones)

| Var                                                                                                          | Purpose                                         |
|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <code>APP_ENV=production</code> , <code>APP_DEBUG=false</code> , <code>APP_KEY</code> , <code>APP_URL</code> | Core (debug <b>must</b> be false in prod)       |
| <code>DB_DATABASE</code> / <code>USERNAME</code> / <code>PASSWORD</code>                                     | MySQL                                           |
| <code>QUEUE_CONNECTION=database</code>                                                                       | Queue worker                                    |
| <code>SANCTUM_STATEFUL_DOMAINS</code>                                                                        | CORS/auth domains                               |
| <code>SEED_ADMIN_PASSWORD</code>                                                                             | Required before seeding <code>UserSeeder</code> |
| <code>GATEWAY_INTERNAL_SECRET</code>                                                                         | Shared backend↔gateway secret                   |
| <code>GATEWAY_BASE_URL</code>                                                                                | Public gateway URL for agent enrollment         |

## 8. Queues

Port scans and **RAG indexing jobs** require a worker:

```
# /etc/systemd/system/quenyx-queue.service (ExecStart)
/usr/bin/php artisan queue:work --sleep=3 --tries=3 --max-time=3600
```

## 9. Scheduler

QynSight checks require cron running the Laravel scheduler:

```
* * * * * cd /path/backend && php artisan schedule:run >> storage/logs/scheduler.log 2>&1
```

Without it, monitoring shows "Last poll: never".

## 10. Cache

`config:cache`, `route:cache`, `view:cache` in production. Use Redis for cache/sessions in multi-node (optional). Run `config:clear` after `.env` changes in dev.

## 11. AI / RAG / Copilot environment (opt-in)

All AI is **off by default**. Enable deliberately:

| Var                                                                                              | Default            | Effect                                                  |
|--------------------------------------------------------------------------------------------------|--------------------|---------------------------------------------------------|
| <code>AI_PROVIDER</code>                                                                         | <code>mock</code>  | <code>openai</code> to use real models                  |
| <code>AI_ENABLED</code>                                                                          | <code>false</code> | master switch for real model calls                      |
| <code>OPENAI_API_KEY</code> / <code>OPENAI_MODEL</code> / <code>OPENAI_EMBEDDINGS_MODEL</code>   | unset              | provider config (models from env only)                  |
| <code>COPILOT_ENABLED</code>                                                                     | <code>true</code>  | Copilot on (still mock unless <code>AI_ENABLED</code> ) |
| <code>AI_PROMPT_LOGGING_ENABLED</code>                                                           | <code>false</code> | store prompt content (keep off)                         |
| <code>AI_CONVERSATION_PERSISTENCE_ENABLED</code>                                                 | <code>false</code> | persist conversations (keep off)                        |
| <code>AI_COPILOT_RAG_ENABLED</code> / <code>RAG_ENABLED</code> / <code>EMBEDDINGS_ENABLED</code> | <code>false</code> | RAG runtime                                             |
| <code>VECTOR_PROVIDER</code>                                                                     | unset              | <code>openai</code> (metadata-only today)               |
| <code>RAG_INDEX_TENANT_EVIDENCE</code>                                                           | <code>false</code> | <b>keep off</b> (privacy)                               |

## 12. Storage

Ensure `backend/storage/app/agents` exists and is writable by the web user (agent binaries + on-demand Go build). Standard Laravel `storage/` and `bootstrap/cache` permissions apply.

## 13. Database & migrations

---

```
php artisan migrate --force
php artisan migrate:status # verify all applied
```

## 14. Seeders & corpus

---

```
php artisan db:seed --force # core seeders (set SEED_ADMIN_PASSWORD)
php artisan db:seed --class=ComplianceCorpusSeeder --force
php artisan compliance:seed-source-documents \
  database/corpus/nca/ecc-2-2024/source-documents.json \
  --framework=nca-ecc --release=2:2024
```

Expected corpus: **5 domains, 108 controls, 108 requirements**, active **Revision v1**.

## 15. Backup & restore

---

- **DB:** `mysqldump` nightly; store offsite/encrypted. Restore: `mysql < dump.sql`.
- **Storage:** back up `backend/storage/` (uploaded artifacts, agent binaries).
- **Config:** keep `.env` in a secrets manager (never in git).

## 16. Rollback

---

- **App:** `git checkout <previous-tag>` → rebuild backend/frontend/gateway → restart services.
- **DB:** restore from the pre-deploy dump. **Do not** rely on `migrate:rollback` for destructive changes in production; restore from backup.
- **Corpus:** roll back to the prior active revision (deterministic snapshots; see Doc 18).

## 17. Health checks

---

- Backend: `GET /api/health`.
- Gateway: `GET /health` → `{"status":"ok","service":"gateway"}` (and `/ready`).
- Monitor ports 8000 (backend) and 4000 (gateway); alert on 502/503 and DB connectivity.

## 18. Production checklist

---

- ☐ `APP_ENV=production`, `APP_DEBUG=false`, strong `APP_KEY`, HTTPS `APP_URL`.
- ☐ `config:cache` + `route:cache` after deploy.
- ☐ CORS restricted to frontend origin; `SANCTUM_STATEFUL_DOMAINS` set.
- ☐ `GATEWAY_INTERNAL_SECRET` set on both sides.
- ☐ `SEED_ADMIN_PASSWORD` set; seeded credentials rotated.

- ☐ Queue worker + scheduler running.
- ☐ AI/RAG flags reviewed (off unless intended; tenant-evidence indexing off).
- ☐ Migrations applied; corpus counts verified.
- ☐ Backups scheduled.